

The van Hiele levels

TCE 318P · Week 3 · Geometry & van Hiele — a reference sheet from Todd, to read alongside Higgins

This page is not part of Higgins's article. It is our own summary, added for this course. The levels are van Hiele's; the wording here is ours. Sources are listed at the bottom.

Level	What the learner is doing	What it sounds like — from Higgins's second graders
1 Visual	Describes a shape by how the whole thing looks , or by how it is turned. A square and a rectangle feel like separate, unrelated things. The name comes from the overall picture, not from any part of it.	<i>"Some shapes are long — squares are short; rectangles are long."</i>
2 Analysis	Describes a shape by its properties and parts — sides, corners, straight or curved, open or closed. The properties are real and used, but each class of shape is still its own separate box.	<i>"All shapes have straight lines except for circles and ovals."</i>
3 Abstraction	Relates the properties to one another and orders the classes logically. One kind of shape can be a special case of another — so a square can be a rectangle without stopping being a square.	<i>"I think they should be in with the other shapes because they do not have sides but are still in the group."</i>

Levels 4 (Deduction) and 5 (Rigor) complete the model — formal proof, and comparing axiom systems. They are named here for completeness and are not in play in this course. Nothing a grade 2–4 learner says should be filed under them.

Three things to hold onto

- **The levels are sequential, invariant and hierarchical.** A learner works through them in order and cannot skip one. Experience moves someone up, not age — an adult who has not thought about shapes since grade school can sit at Level 1.
- **Locate, don't rank.** You name a level in order to offer the right next experience. A "diamond" answer is data, not an error. An experience aimed above where someone is only gets memorised.
- **The label tells you almost nothing; the justification is the diagnosis.** Two learners can both say "rectangle" from completely different levels. Always ask *how do you know?*

A note on the numbering

You will see this ladder numbered two different ways, and both are correct. Van Hiele originally numbered the levels **0–4**. American writers renumbered them **1–5**, freeing up "level 0" to describe a child who does not yet recognise shapes at all. Higgins explains this on p. 493 of the article you just read. **In this course we use 1–5** — so Visual is Level 1, and what some texts call "Level 0" or "informal deduction" you will find here as Levels 1 and 3.

Sources. The three-level table is our own wording, built from:

Higgins, K. M. (2013). Building a team of learners. *Teaching Children Mathematics*, 19(8), 490–496. Student quotations above are from pp. 491–492; the note on numbering is from p. 493.

Howse, T. D., & Howse, M. E. (2015). Linking the Van Hiele theory to instruction. *Teaching Children Mathematics*, 21(5), 304–313. The five-level table appears on p. 306.

Clements, D. H., & Battista, M. T. (1992). Geometry and spatial reasoning. In D. A. Grouws (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 420–464). Macmillan.

van Hiele, P. M. (1999). Developing geometric thinking through activities that begin with play. *Teaching Children Mathematics*, 5(6), 310–316.